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DESIGN COMPUTATION & DIGITAL ENGINEERING LAB

TOPODIFF

Diffusion Models Beat GANs on Topology Optimization

François Mazé[1], Faez Ahmed[1]
[1]MIT

THE BASICS

TopoDiff introduces diffusion models to structural topology optimization. It uses a new guidance strategy to make diffusion models performance-aware and constraints-aware. In this work, we present three datasets for data-driven topology optimization and introduce a diffusion model incorporating external guidance, which outperforms a state-of-art GAN. You can find below the most up-to-date information on this project:

Paper

Code

About the dataset

Download the dataset

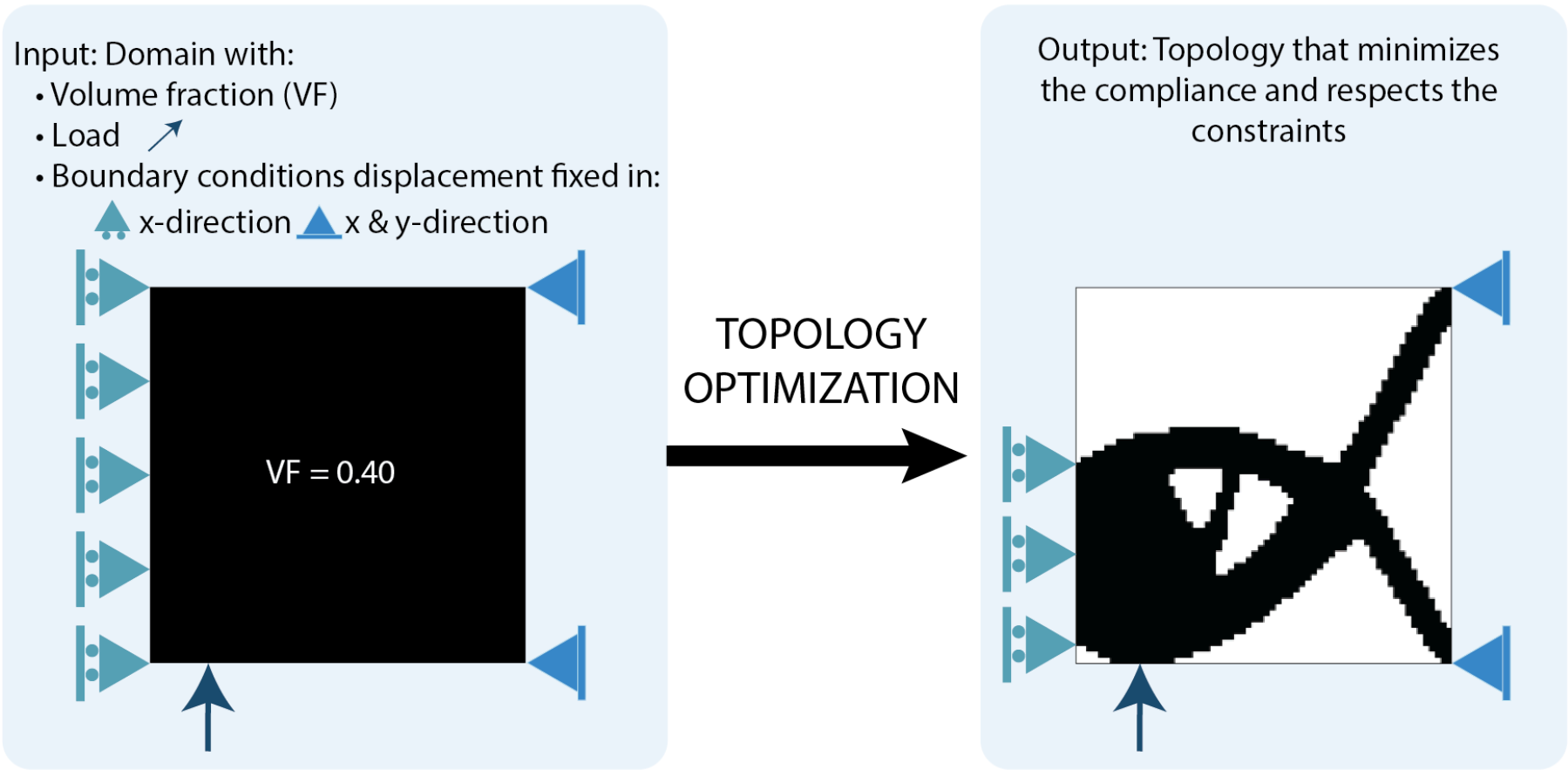
Download the trained model

Video Presentation

AAAI poster

TOPOLOGY OPTIMIZATION

Topology optimization (TO) aims to find an optimal subset of material included in the full design domain under a set of displacement boundary conditions and loads applied on the nodes of the domain and a volume fraction condition. A structure is optimal when it minimizes an objective function, such as mechanical compliance, subject to constraints.

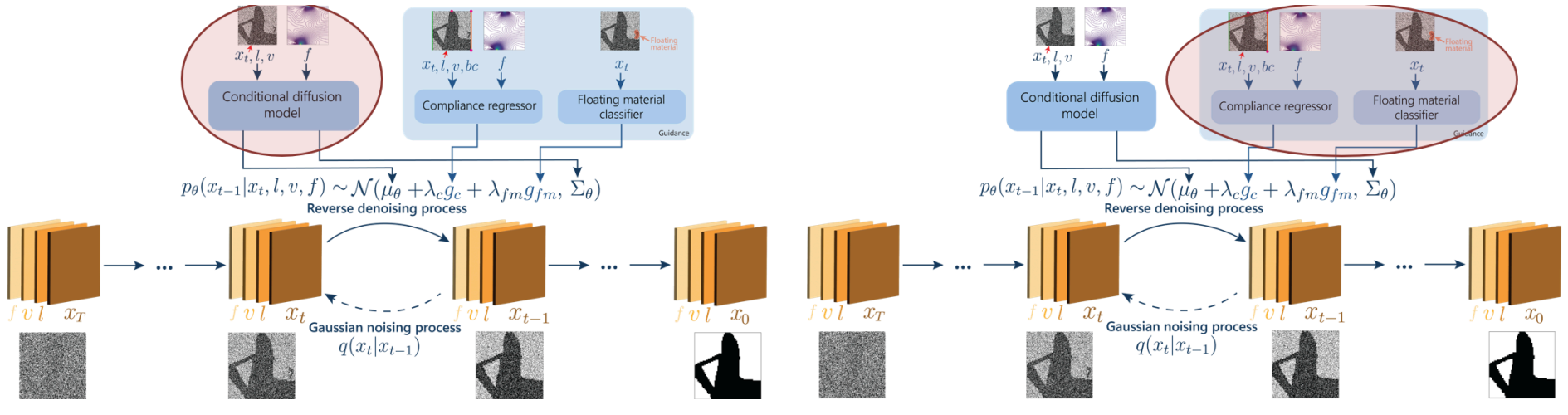


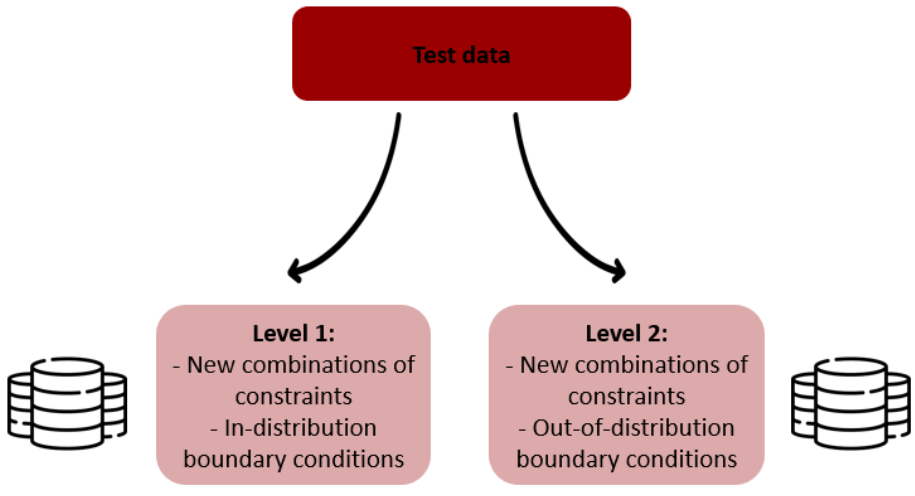
Traditional TO methods rely on Finite Elements Analysis (FEA). One popular method is the Solid Isotropic Material with Penalization (SIMP) method. SIMP associates every element of the mesh with a continuous density to perform gradient-based methods. These traditional methods have two major pitfalls : they are often slow, and may lead to sub-optimal solutions because of the use of penalization factors or post-processing.

Many deep learning methods have recently been developed to speed up and improve the quality of topology generation. The most recent works rely on conditional GANs to generate new topologies based on input constraints. Yet promising, most of these methods show limited generalization ability to unseen constraints and often generate manufacturable designs.

OUR DATASET

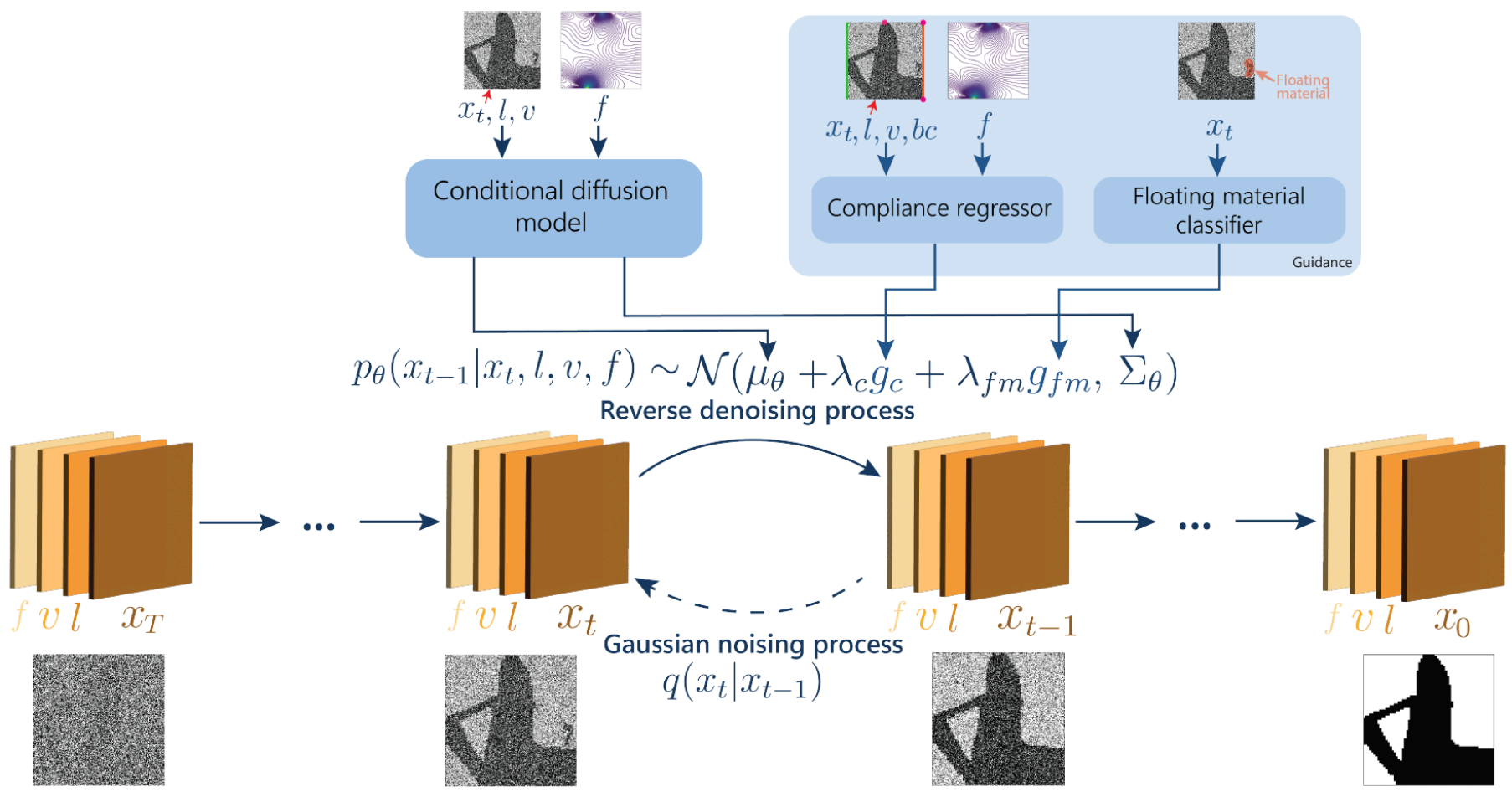
Discover below how TopoDiff's dataset was built. More details are provided on the Dropbox containing the downloadable dataset.





OUR DIFFUSION MODEL ARCHITECTURE

TopoDiff's main architecture is a conditional diffusion model, which takes as input a randomly sampled white noise and the design constraints and gradually denoises the noise until reaching an actual mechanical structure. In order to improve the performance and manufacturability of the generated structures, we introduced a guidance mechanism to encourage denoising steps that minimize the compliance predicted by a separate regressor and avoid floating material, detected by a separate classifier. The figure below summarizes TopoDiff's architecture:



EVALUATION OF PERFORMANCE

We benchmarked our model with TopologyGAN, a state-of-art cGAN architecture for topology optimization, using four metrics assessing the physical performance of the structures, but also the manufacturability and respect of constraints. Our model outperforms TopologyGAN on all metrics of interest. We also conducted an ablation study to evaluate the impact of our guidance strategy on performance.

	Level 1 test data			*Level 2* test data		
Model	TopologyGAN	Unguided TopoDiff	Guided TopoDiff	TopologyGAN	Unguided TopoDiff	Guided TopoDiff
Average Compliance Error (%)	48.51 ± 16.38	**4.10** ± 0.88	4.39 ± 0.94	143.08 ± 38.50	22.13 ± 8.52	**18.40** ± 5.88
Median Compliance Error (%)	2.06	**0.80**	0.83	6.82	1.88	**1.82**
Proportion of Compliance Error >30% (%)	10.11	**2.33**	2.56	24.10	8.20	**8.10**
Average Volume Fraction Error (%)	11.87 ± 0.52	1.86 ± 0.03	**1.85** ± 0.03	14.31 ± 0.75	1.81 ± 0.04	**1.80** ± 0.04
Proportion of Load Violation (%)	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Proportion of Floating Material (%)	46.78	6.64	**5.54**	67.90	7.53	**6.21**

Table 1: Comparison of performance between TopologyGAN and TopoDiff (guided and not guided) on the two level test sets. Values after ± indicate the 95 % confidence interval around averages. The values in bold are the best ones for each level.

CITATIONS

CHICAGO

François Mazé, and Faez Ahmed. "Diffusion Models Beat GANs on Topology Optimization" In Proceedings of the AAAI Conference on Artificial Intelligence, 2023.

BIBTEX

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CONTACT

MIT Building 3, Room 3-470
33 Massachusetts Ave
Cambridge, MA 02139

faez@mit.edu

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